PLMC

PRODUCT LIFECYCLE MANAGEMENT CENTER
AT CLEMSON UNIVERSITY

2024-2025 Annual Report

for the
Product Lifecycle Management Center at Clemson University

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Contributing

John Wagner, PhD, PE
Professor of Mechanical Engineering
Director of PLM Center

212 Fluor Daniel EIB
Clemson University
Clemson, South Carolina
29634-0921 USA

(864) 656-7376
jwagner@clemson.edu

John Morris
PLM Applications Engineer

328A Fluor Daniel EIB
Clemson University
Clemson, South Carolina
29634-0921 USA

(801) 850-3126
jhmrrs@clemson.edu

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Executive Summary

During the 2024-2025AY the PLM Center made strides in conducting digital engineering research, teaching about PLM topics, training students on PLM software, and supporting projects and classroom activities. A quantification of the center's outreach is shown in Table 1.

	Outreach to Students/Faculty	PLM Process Workshops		PLM Software Training Workshops		PLM Research Projects		
Year	Persons	Events	Persons	Events	Persons	Events	Persons	**Total**
2019	166			1	10			176
2020	448	3	123	7	119	2	2	692
2021	619	1	30	3	34	2	2	685
2022	380	14	71	8	70	3	3	525
2023	800	20	102	6	71	3	3	976
2024	527	16	110	2	39	4	2	678
2025	**674**	**16**	**96**	**16**	**111**	**5**	**3**	**884**

Table 1: Number of events hosted by the PLM Center and attending individuals by year.

Though the live seminars and workshops diminished in attendance, the PLM Center successfully increased delivery through its on-demand learning resources available on its Canvas workgroup page. The PLM Center has begun championing software certification efforts in partnership with the department of General Engineering. Certifications in software from three different vendors are available through the PLM Center, with 21 students becoming certified during the latest academic year.

Assistance and training continue to be provided to students in various research projects, such as student engineering teams, senior design, and semester projects. This included development of a modeling kernel enabling declarative simulation of generalized systems. The PLM Center led 18 students in an undergraduate research project creating a digital twin of a robotic vehicle.

The Center continues to work towards building partnerships with various PLM vendors, engaging in discussions and conducting research studies with Siemens, Ansys, PTC, MathWorks, and Dassault Systèmes. Dr. Satchit Ramnath leads the effort in advancing the center's industrial consortium.

1. Education

The education efforts of the PLM Center have focused on fulfilling the mission statement of providing educational tools to students, staff, and faculty at Clemson University. The motivation of these efforts lies in the importance of learning about digital engineering in contributing to modern industrial practices. To these ends the PLM Center has developed a dual teaching and experiential approach for both informing students about concepts as well as providing them hands-on experiences with PLM tools.

1.1. Seminars and Workshops

The PLM Center has put significant effort into curating a set of training modules teaching digital engineering, PLM concepts, and technical skills. A list of these modules is given in Table 2. The modules are divided into four general categories:

1. Fundamental Concepts, introducing PLM and its applications.
2. Digital Product Development, describing digital engineering activities during product design.
3. Graphical Concepts, covering functions generally performed using graphical software applications.
4. Non-Graphical Concepts, related to process and organizational based topics such as project management and systems engineering.

PLM Index	Title
1.1	Introduction to Product Lifecycle
1.2	Digital Components of PLM
1.3	PLM Functional Areas
1.4	PLM in Practice
2.1	Product Design Tools and Approaches
2.2	Data Preservation and Security
2.3	Additive Manufacturing (including 3D printing)
2.4	Data Analytics and Mining
3.1	Computer Aided Design Methods (CAD)
3.2	Computer Aided Engineering (CAE)
3.3	Two Dimensional Drawings and GD/T
3.4	System Simulation and Optimization
4.1	Project Management
4.2	Product Data Management
4.3	Engineering Change Management
4.4	Model Based Systems Engineering (MSBE)
	NX Training
	Teamcenter Training

Table 2: List of training modules and workshops developed by the PLM Center.

Seminars on these topics have been provided since the PLM Center was first established. These hour-long learning opportunities are attended by undergraduate and graduate students in various departments but primarily the Department of Automotive Engineering, Department of Mechanical Engineering, and General Engineering. The PLM Center Computer Laboratory in 352 Fluor Daniel is fully operational; similarly, classrooms on the CU ICAR campus are regularly used for

instruction. In many instances, students remotely accessed the sessions (zoom) and software (CU APPS/Citrix) from their own laptop or desktop computers. Note that attendance at these weekly workshops trended toward smaller numbers as the year progressed. Attendance numbers, which were down possibly due to lack of advertising presence, are given in Table 1.

The teaching effectiveness survey for the workshops and training sessions (refer to Appendix G) was approved by the Clemson University Institutional Review Board (IRB). During the 2023-2024 AY, this survey was deployed to participants following all training sessions, though an insignificant number of responses were collected.

1.2. On-demand Learning

The PLM Center has put significant energy into making these seminars available on-demand. This is accomplished via a website hosted on the Learning Management System Canvas as a workgroup page. This project was initiated during the 2023-2024 academic year and represents a dynamic repository of information, tutorials, and guides for students. Over 700 students joined the Canvas course over the 2024-2025 AY, giving them access to all modules. The Canvas webpage also provides users information about the PLM Center, available resources, and contact information. The Canvas page is broken into three focuses, shown in the following sections.

1.2.1. Seminars and workshops

This site focus, shown in Figure 1: Views of seminars and workshops listed on Canvas (left); and view of individual module with recording and related references (right).Figure 1, is where past presentations are listed and made accessible for viewing. Pages for each specific module have also been developed showing the latest recording as well as related information, references, and links of topics for further investigation.

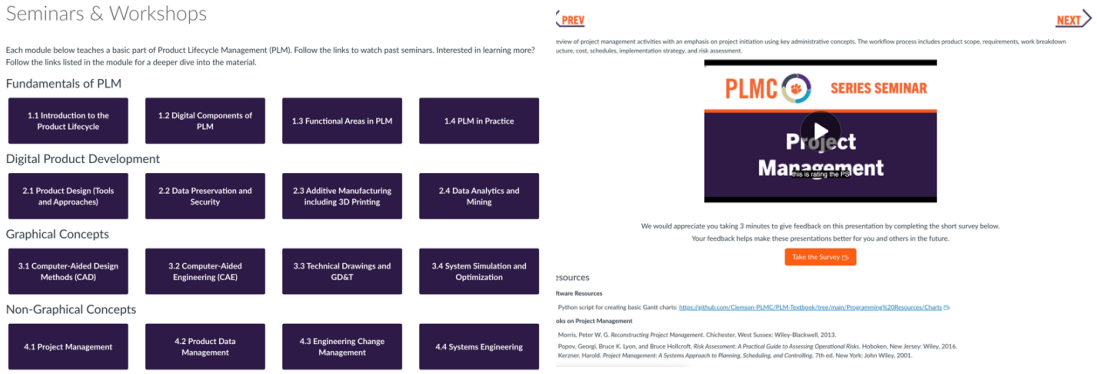

Figure 1: Views of seminars and workshops listed on Canvas (left); and view of individual module with recording and related references (right).

1.2.2. Software Resources

This section overviews various software applications and platforms used in digital engineering. This includes CAx software such as SOLIDWORKS, NX, Onshape, and Ansys, as well as scripting platforms like MATLAB, Simulink, and Arduino and languages such as Python. Each page provides tutorials for how to use the software, links to references and documentation, and in

some cases guided trainings provided on-demand by the PLM Center. The site also provides access information curated for Clemson students, such as which packages are available at locations. Finally, the site provides resources for certification. Screenshots of these pages with resources are shown in Figure 2.

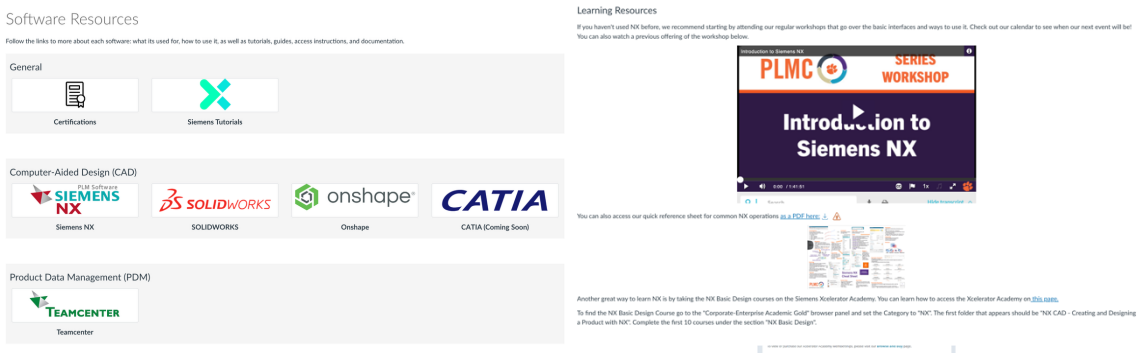

Figure 2: Screenshots from the Canvas page showing some of the software resource pages available (left); and a selection of the guide to Siemens NX (right).

The high value computer packages in the PLM Center may be accessed after Clemson University students have completed a short on-line training program that emphasizes best software practices. The creation of this training was championed by Trae King (CECAS CORE); the materials are hosted on the Bridge training platform. This training is required for students accessing software resources on the Canvas page.

1.3. Certification

During the 2023-2024 AY, the PLM Center partnered with the Department of General Engineering to provide certifications for SOLIDWORKS using PLM equipment. The Canvas page helps students to sign up for certification exams (using vouchers made available by Dassault Systèmes) and provides study guides and exam requirements. Students are able to take the certification exam using workstations in the PLM Center Computer Laboratory.

During the current reporting period, 21 students were able to take their examinations in the computer laboratory. Each certification officially certificates the examinee of their technical understanding of the program. Certifications are tracked by Dassault Systèmes and can be listed on the student's resume, greatly increasing their employability.

Certifications are also freely provided for products owned by Ansys and Siemens, though no students have yet sought certification from these vendors. Note that although Siemens offers free vouchers for academic certifications, they have currently advised academic partners to refrain from certification for the time being until they have improved their examinations.

1.4. Classroom Learning

The PLM Center previously supported the creation of a new senior-level elective course offered for the first time during the Fall 2023 semester. 30 undergraduate and 11 graduate students in the Mechanical Engineering departments enrolled in the course, which provided an in-depth overview

of many of the PLM concepts shown in Table 2. Students in the course gained technical skills in CAD (including using the Siemens NX application), simulation, 3D printing, and drafting, as well as learning Model-Based Systems Engineering (MBSE), project management, and the role of PLM in modern industry.

Supporting the new course was the drafting of a textbook: “Product Lifecycle Management: Processes for Modern Industry,” with over 250 pages of text, figures, tables, and exercises on PLM topics, as well as an original project detailing the design, production, management of a handheld pumping device. The table of contents for this textbook is provided in Appendix H, with some supplementary pages shown in Figure 3.

The Department of Automotive Engineering courses AuE 4080/6080 (Vehicle Testing & Characterization), AuE 8830 (Automotive Manufacturing), AuE 8860 (Vehicle Noise), AuE 8820 (Systems Integration Concepts and Methods), and AuE 8830 (Applied Systems Integration) used PLM software for instruction. The Department of Mechanical Engineering sophomore seminar, ME 2000, and senior capstone design course, ME 4020, received presentations to encourage student use of PLM software. Finally, the technical elective courses ME 4170/6170 (Mechatronics System Design) used PLM software in the classroom.

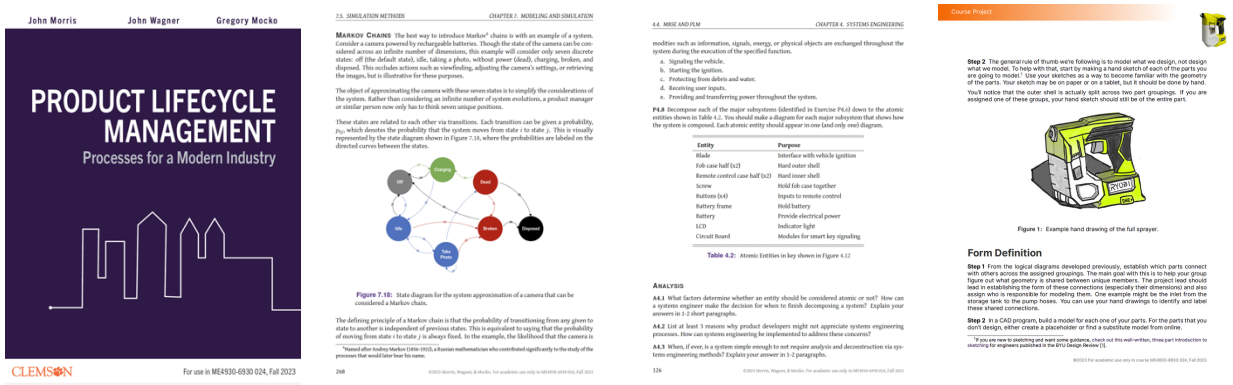

Figure 3: Pages from unpublished textbook on PLM (from left): cover; overview of Markov chains; exercises on systems engineering; project involving sketching the pumping mechanism.

2. Research and Industrial Consortium

2.1. Guided Projects

Research projects begun in the PLM Center have resulted in the graduation of three students (all in the department of Mechanical Engineering): two master’s thesis students (D. Procopio, M. Kale) and one undergraduate student (F. Rowell) who finished an ME H3150 honors research. Mr. Rowell has continued this research with the PLM Center as a master’s student. Additionally, a PhD student in Mechanical Engineering (J. Morris) proposed a dissertation at the end of the 2023-2024AY focusing on digital twin representation that is expected to continue throughout the 2024-2025AY.

An HVAC (chiller plant, boiler, air handling) collaboration with the Clemson University Facilities Team and Optimum Energy LLC (provider of real time monitoring software) has temporarily halted. An additional project creating a digital twin of a centrifugal pump used in the chiller plant refrigeration waste heat cycle was initiated near the end of the 2024 academic year.

The use of Product Data Management (PDM) in collaborative design teams is being studied in an ongoing project to understand how PLM affects engineering processes. This specific study is evaluating the impact that the use of Siemens Teamcenter has on student design groups, and is being performed in connection with the ME 4020 capstone design course.

The PLM Center guided the development of open-source software tools that enable declarative simulation of generalized systems. The software, known as ConstraintHg, is available from GitHub and from the Python Package Index (PyPI) under the MIT license.

2.2. Supported Projects

The PLM Center has engaged with multiple student engineering teams to support their technology and fundamental processes. The Clemson Rocket Engineering club met with the PLM Applications Engineer and attended multiple group training sessions. The PLM Center also worked with Ansys Inc. to provide analytic software for the Clemson Formula SAE student team. More recently, a Department of Industrial Engineering graduate student has begun using PLM software (Siemens NX) for their PhD thesis.

The US Army-Clemson University VIPR-GS (Virtual Prototyping of Ground Systems) has created a digital twin for off-road tracked autonomous vehicles (using MATLAB/Simulink/Simscape) and support virtual collaboration on the Deep Orange 14 Vehicle Design. This has continued with the the DO 15, 16, and 17 Teams. The center additionally provides resources and training for over 20 researchers using PLM software under the VIPR-GS center.

2.3. Undergraduate Research

For eight continuous semesters, the PLM Center has directed an undergraduate research project sponsored by the Clemson Creative Inquiry program. This project brings in a group of undergraduates from various STEM majors with a goal of providing experiences using PLM tools while developing a digital twin of a tracked, scaled, robotic vehicle. Participation in this activity was eight students in Fall and sixteen in the Spring semesters. Students learned Siemens NX, performed dynamic simulation using MATLAB and Python, constructed sensors using integrated circuits, and used Arduino microcontrollers to host bespoke IoT platforms. The students presented their efforts in the 2025 Focus on Creative Inquiry Poster Session.

2.4. Industrial Consortium

The PLM Center welcomed Dr. Satchit Ramnath aboard as a faculty partner leading the expansion of the Industrial Consortium in 2023. The Industrial Consortium has been previously impacted by the COVID-19 pandemic. The PLM Center governance documents are completed and await designated Clemson University official signature.

3. Service and Outreach

The PLM Center at Clemson University has participated in outreach with South Carolina industry, the Clemson University Facilities Group, and the Pickens County Career & Technology Center. These entities have provided opportunities for student engagement and research projects, as well as outreach to companies. The South Carolina Manufacturing Extension Partnership (SCMEP) has been engaged to survey companies across the State of South Carolina regarding their use of digital technology including PLM software and Digital Twins. GE Power collaborated on a graduate student research project examining model change updates in the CAE NX software for gas turbine turbulators (2019-2021). In addition, GE Power was the inaugural member of the Industrial Consortium before the COVID-19 pandemic.

The Clemson University Facilities Group collaborated on a project (2020-2021) to create a digital twin for the East Side Chiller Plant for improved performance. A new project commenced (2022-2024) with the investigation of a "green power" HVAC system (electric boiler, AC) using a wind turbine microgrid to heat and cool a commercial building. Through periodic interaction with the local school district, Pickens County, Career & Technology Center, we maintain visibility of their student's use of CAD software for CNC machining.

Learning materials have been created in support of the Clemson University PEER & WISE (Programs for Educational Enrichment and Retention; Women in Science and Engineering) Summer Camps Program for middle school students. Multiple day (75-minute sessions) of lecture and hands-on activities were prepared on the topic of "Mechanical Engineering Theory, CAD Drawing, and Geared Transmission Systems" using hands-on kits, SolidWorks, etc. Over several summers, K-12 students have participated in sessions located in Fluor Daniel EIB and the PLM Center Computer Laboratory.

The PLM Center maintains an emailing list of 209 individuals used to periodically advertise digital engineering events happening on Clemson campus such as vendor training sessions, seminars, or competitions.

The PLM Center also offered training events for NX, SOLIDWORKS, and Teamcenter, hosted training by three representatives from TTI LLC for Ansys Fluent, and facilitated virtual training sessions between Siemens, Ansys, and Mathworks representatives with student clubs, ME capstone teams, and faculty.

4. Software Partnerships

The donation of licenses to Clemson University by PLM Software Vendors is a core foundational element of the PLM Center. These academic-industry partnerships need to be nurtured through collaborative efforts to ensure that the software packages are actively used for education, research, and service as outlined in the governance documents.

4.1. Siemens

Siemens Digital Industries Software provides FactoryCAD, NX, Simcenter (AMESIM), Teamcenter and other software packages including technical services. The donated software is actively used across the Clemson University campuses (Clemson, SC, Greenville, SC, Charleston,

SC) for courses and research projects. During the Spring 2025 semesters, Teamcenter was actively used in ME 4150 (Senior Honors Thesis), NX was used in ME 4710/6710 (PLM Software) and both software platforms were employed by the Deep Orange Teams for classroom and vehicle design activities.

The Siemens software is supported by their online Xcelerator Academy learning platform. All students, faculty, and staff can be provided access to the training content through the PLM Center via the CECAS CORE team (Mr. Trae King). The platform provides self-paced tutorials for the Siemens NX (CAD, CAM) and Teamcenter applications as well as additional training through their resource center. The Siemens NX basic features and Xcelerator Academy (Learning Advantage) setup were placed into a self-guided presentation by the PLM Center. Furthermore, Siemens has diligently provided onsite campus training and online technical support for their software packages throughout the academic year. As led by CECAS CORE, the hosting for Teamcenter is planned to shift from university-owned servers to space rented from Amazon Web Services, which will provide better functionality and service than the aging infrastructure.

4.2. Dassault Systèmes

Dassault Systèmes previously provided the complete 3D Experience online PLM platform which included CATIA, DELMIA, ENOVIA, and SIMULIA. Some of these software packages were introduced into the technical elective course ME 4710 (Advanced Computer Aided Engineering and Design) when available. Students were required to design and analyze their design projects using these tools. Dassault Systèmes offered online training video(s) which were effective in getting students started with 3D Experience.

Over the 2019 Summer, Dassault Systèmes (F. Salako) provide three days of onboarding training for the 3D Experience. Participates learned how to use the new platform and create engineering projects. In the 2020 Spring, they brought the “Simulation on Wheels” to Clemson main campus to demonstrate their software applications. This event had strong participation with participants immersed in simulation, virtual reality, and augmented reality applications. However, the visit ended early due to the COVID-19 virus and corporate decision to cease travel. Currently, no Dassault Systèmes software is widely available on campus except for SOLIDWORKS. Dr. S. Ramnath obtained the use of three seats of CATIA for use in on campus research projects in partnership with Honda.

Recent discussions with the Dassault Team (19 September 2022, Summer 2023) have focused on the addition of further CATIA software seats for the Department of Automotive Engineering and PLM Center to support Model Based Systems Engineering (MBSE) activities.

5. Center Organization and Structure

5.1. Infrastructure and Locations

PLM Center Computer Laboratory: 352 Fluor Daniel EIB[1]

[1] The PLM Center Computer Laboratory was located in 352 Fluor Daniel EIB for the entirety of the reporting period. At the start of the 2025-2026AY, the laboratory was relocated to room 260.

The PLM Center offers a space for students to collaborate on their virtual design and research projects. With each PLM Overview presentation, students are invited to request access to the laboratory which is controlled through electronic cards. Students have card access to 28 high end workstations 24 hours per day, 7 days a week. Each workstation has the latest software from Siemens and Dassault Systèmes. Students are provided with training programs to begin learning to use these software packages. This training is explained in more detail in the software partnerships section of this report. The center also features white boards and project for student to practice presentations and discuss their projects.

Physical access to 352 EIB is electronically controlled by a door lock; Mr. Austin Breazeale in the ME Machine Shop (lower level of Fluor Daniel) provides access to students upon written request. Students may request access electronically via a form provided on the center's website.

Starting in Summer of 2025, the PLM Center Computer Laboratory was relocated to the second floor of the Fluor Daniel Building (room 260). Additionally, several of the workstations were moved to the high bay project area in Fluor Daniel to provide access to engineering students engaged in senior design and other projects. Functionality, resource provision, and available software were not affected by this move, which was completed by the start of the 2025-2026 AY.

PLM Applications Engineer Office: 328A Fluor Daniel EIB

Website: https://www.clemson.edu/centers-institutes/plm/index.html

Email address: plmcenter@clemson.edu

Canvas: Workgroup page entitled "PLM Center" (hosted by jwagner@clemson.edu)

- Link to enroll (available for university personnel): https://clemson.instructure.com/enroll/AMDEFW

5.2. PLM Center Legal Governance Documents

The PLM Center Director continued to work with Dr. Anand Gramopadhye (CECAS Dean), Ann Marie Alexander (Assistant Vice President, Strategic Corporate Partnerships) and Clay Grayson (Grayson Law Firm, Mt. Pleasant, SC) at the request of Clemson University Office of General Counsel (Chip Hood) to revise the PLM Center governing documents. After extensive revisions, these four governing documents have been completed and await Clemson University signature:

- PLM Center Charter
- PLM Participant Agreement – Joinder Agreement
- Participant Designation Chart
- Policy Regarding Participant-Involved Scientific Research Projects

Siemens Digital Industries used the existing Software License & Services Agreement (#60035586) and "Software License Clarification Letter" (dated 22 February 2022) for the Clemson University partnership which now lists education and research opportunities within the PLM Center context.

The focus will shift to Dassault Systèmes with the governance documents. Discussion have occurred with Dassault Systèmes, the Department of Automotive Engineering, and the PLM Center regarding the use of CATIA for Model-Based Systems Engineering (MBSE).

5.3. Personnel

The PLM Center is led by Professor John Wagner working closely with the PLM Applications Engineer, John Morris, to create learning materials, deliver educational programming, guide research, and provide campus outreach. Dr. Satchit Ramnath serves as the associate director of research, focusing on creating valuable partnerships with external industry organizations. In addition, Dr. Greg Mocko provides assistance with advanced engineering design topics and research guidance. The graduate level research projects by MSME and PhD students are within the Department of Mechanical Engineering. Recently, the Deep Orange (DO) Vehicle Teams on the CU ICAR Campus have begun using Siemens NX and Teamcenter which offers new outreach activities.

To complete the Governance Documents, an appointed Clemson University Leader must sign off on them. Over the past several years, we have worked with Dr. Anand Gramopadhye (Dean, College of Engineering, Applied Science, and Computing), Ann Marie Alexander (Office of Corporate Partnership), Attorney Clay Grayson (external), and Siemens.

The financial materials have been handled by the accountant(s) in the Department of Mechanical Engineering and CECAS Business Office. Updates to the PLM Center at Clemson University website were completed by Kimberly Williams, CECAS PROMO.

The official personnel associated with the center is as follows:

Director – John Wagner, PhD, PE
Professor, Department of Mechanical Engineering
212 Fluor Daniel EIB, 864-656-7376, jwagner@clemson.edu

Associate Director of Education and Training – Greg Mocko, PhD
Associate Professor, Department of Mechanical Engineering
243 Fluor Daniel Building, 864-656-1812, gmocko@clemson.edu

Associate Director of Research – Satchit Ramnath, PhD
Assistant Professor, Department of Mechanical Engineering
210 Fluor Daniel Building, 864-656-6248, sramnat@clemson.edu

PLM Applications Engineer – John Morris
PhD candidate, Department of Mechanical Engineering
328A Fluor Daniel EIB, 801-850-3126, jhmrrs@clemson.edu

Industrial Partnerships – Ann Marie Alexander
Assistant Vice President, Office of Corporate Partnerships and Strategic Initiatives
405 S. Palmetto Blvd – Office 439A, 864-506-6628, amalex@clemson.edu

5.3.1. PLM Applications Engineers

In the Fall 2018, a nationwide search was conducted to hire a PLM Applications Engineer. In March 2019, Eric Lambert left ABB (Greenville, SC) and joined Clemson University as a Lecturer in the Department of Mechanical Engineering with full time responsibility to the PLM Center. He helped to launch the Center over a 13-month period before departing for GE Power (Greenville, SC) in April 2020 to be an Additive Operations Engineer.

In August 2020, Chengshi (Charles) Wang was hired as the Applications Engineer for 10 hrs/wk while also teaching ME 3070 (Foundations of Mechanical Systems) in the Department of Mechanical Engineering. We appreciate the cooperation of Dr. Atul Kelkar, Mechanical Engineering Department Chair, in this joint student lecturer appointment. He graduated in December 2020 and joined Argonne National Laboratory (Chicago, IL) as a Post Doc.

In June 2021, John Morris joined the PLM Center as the third Applications Engineer while studying for his PhD degree. Stipend and tuition waiver support will come from the US Army – Clemson University VIPR GS with the Education, Diversity & Inclusion Initiative to share digital engineering processes and tools with the students and faculty working on the virtual design of ground vehicles. Mr. Morris is regularly offering PLM training sessions and workshops on the Clemson, SC and Greenville, SC campuses. In addition, he will help lead the Creative Inquiry undergraduate research project while pursuing a doctoral degree with Dr. Greg Mocko and Dr. John Wagner with a research project exploring digitalization technology in small-medium enterprises. His expected graduation date from Clemson is December 2025.

6. Appendices

Appendix A. Available Software Packages

A listing of the donated academic and commercial software packages as championed by the PLM Center at Clemson University.

Siemens PLM Software (excess of 500 licenses, donated)

- FEMAP with NX Nastran (1)
- Fibersim Legacy (50)
- NX Academic – Core & CAD (156); NX Academic – CAE & CAM (312)
- Simcenter AMESim (70)
- Simcenter Test Lab (20)
- Teamcenter (200)
- Tecnomatix Manufacturing / FactoryCAD (82)

Dassault Systèmes PLM software

- SOLIDWORKS (site license handled by CCIT for all Clemson University students)
  - Introductory classes offered through General Engineering
  - Certifications championed by John Minor (Professor, General Engineering) and facilitated through PLM Center
- 3DS Experience – including CATIA (previously held)

  Champion: Rob Krulac
  Senior Business Development Director
  Partnership Office
  5 Research Drive
  Greenville, SC 29607
  (864) 283-7126, rkrulac@clemson.edu
- CATIA commercial license (expired – not renewed by CECAS)

PTC PLM Software

- Onshape available for students with academic license, supported by the PLM Center
  - Enterprise edition sponsored by PTC (1)

ANSYS Simulation Software

- Site license handled by CCIT for all Clemson University students
  - Workbench
  - Mechanical
  - Fluent
  - Icepack
- Commercial packages sponsored by Ansys (1)
- Training provided by Ansys and supported by PLMC

Mathworks Software for Engineered Systems

- Site licenses handled by CCIT for all Clemson University students

    - MATLAB
    - Simulink
    - Simscape
- Training provided by Mathworks and supported by PLMC

Appendix B. Faculty Use of PLM Center Championed Software

Assessment mapping of donated software in terms of faculty use in classes as well as research projects on the three Clemson University campuses in South Carolina.

Course/Project	Date	Instructors	PLM Software	Students
AMFG 6800: Practicum Experience in Adv Manufacturing	S24	Matthew Krugh	Simatic	10
AuE 4080/6080: Vehicle Testing & Characterization	F22	Jeff Poland	Simcenter, Testlab	37
AuE 4080/6080	F23	Jeff Poland	Simcenter, Testlab	30
AuE 4080/6080	S25	Matthias Schmid	MATLAB, Testcenter	20
AuE 4620/6620: Digital Manufacturing	S24	Matthew Krugh	SIMATIC	41
AuE 4600/6600: Vehicle Dynamics	F24	Matthias Schmid	MATLAB/Simulink	50
AuE 4860/6860: Digital Automotive Manufacturing	S19	Laine Mears	Mindsphere	
AuE 8010 DO Internships	S22	Chris Paredis	NX, Teamcenter	14
AuE DO10	Su18	Jeff Poland	SCADAS DAQ	5
AuE DO11	S19	Chris Paredis	NX	36
AuE DO11	F19	Chris Paredis	NX	34
AuE DO11	Su20	Chris Paredis	NX	31
AuE DO12	Su20	Chris Paredis	NX	5
AuE DO11	F20	Chris Paredis	NX	31
AuE DO15	F23	James Forbes	NX, Teamcenter	19
AuE DO16	Su24	James Forbes	NX, Teamcenter	22
AuE 8110: Automotive Systems Overview	F19	Chris Paredis	NX	129
AuE 8110	F20	Chris Paredis	NX	48
AuE 8330: Automotive Manu. Process Dev., Methods, and Tools	F17	L. Mears, J. Schulte	Tecnomatix, FactoryCAD	74
AuE 8330	F18	L. Mears, F. A. Niaki	FactoryCAD	72
AuE 8330	F19	Laine Mears	FactoryCAD	42
AuE 8330	F20	Laine Mears	FactoryCAD	31
AuE 8330	F21	Laine Mears	Tecnomatix	37
AuE 8330	F22	Matthew Krugh	Tecnomatix	53
AuE 8330	F23	Matthew Krugh	Tecnomatix	65

AuE 8810: Automotive Systems – An Integrated Overview	F21	C. Paredis, W. Schmueser	NX	127
AuE 8810	F22	C. Paredis, W. Schmueser	NX	106
AuE 8810	F23	James Forbes	NX	6
AuE 8810	F24	James Forbes	NX	17
AuE 8820: Systems Integration Concepts and Methods	S20	Chris Paredis	NX	49
AuE 8820	S21	Chris Paredis	NX	22
AuE 8820	S22	Chris Paredis	NX, Teamcenter	30
AuE 8820	S23	Chris Paredis	NX	17
AuE 8820	S23	Rob Prucka	NX, Teamcenter	31
AuE 8820	S24	James Forbes	NX, Teamcenter	24
AuE 8820	S25	James Forbes	NX, Teamcenter, Amesim, STAR CCM+, MotionView, MATLAB	21
AuE 8830: Applied Systems Integration	F20	Chris Paredis	NX, Teamcenter	41
AuE 8830	F21	Chris Paredis	NX, Teamcenter	16
AuE 8830	F22	Chris Paredis	NX, Teamcenter	24
AuE 8830	F23	Rob Prucka	NX, Teamcenter	31
AuE 8830	S24	Rob Prucka	NX, Testlab	31
AuE 8860: Vehicle Noise, Vibration and Harness	S18	Jeff Poland	Simcenter, Testlab	17
AuE 8860	S19	Jeff Poland	Test Lab	20
AuE 8860	S20	Jeff Poland	Test Lab	30
AuE 8860	S21	Jeff Poland	Test Lab	16
AuE 8860	S22	Jeff Poland	Test Lab	20
AuE 8860	S23	Jeff Poland	Test Lab	27
AuE 8860	S24	Jeff Poland	Test Lab	3
AuE 8870: Methods for Vehicle Testing	F18	Jeff Poland	Test.Lab	35
AuE 8870	F19	Jeff Poland	Test.Lab	35
AuE 8930: Advanced Composites Structures		Srikanth Pilla	Fibersim	TBD
Training Session: LMS Modal Analysis	Su18	Jeff Poland w/Siemens	LMS Modal Analysis	75
CE 8930: Inelastic Materials Modeling and Application	S19	Qiushi Chen	Star CCM+	15

CE 8890: Doctoral Dissertation Research	S19	Quishi Chen	Star CCM+	4
IE 4650/6650: Facilities Planning & Design	S23	Mary Beth Kurz	Tecnomatix	15
ME 2900/3900/4900 PLM Processes Creative Inquiry	F21	J. Wagner, G. Mocko	Siemens NX	7
ME 2900/3900/4900	S22	J. Wagner, G. Mocko	NX	5
ME 2900/3900/4900	F22	J. Wagner, G. Mocko	NX, Teamcenter	4
ME 2900/3900/4900	S23	J. Wagner, G. Mocko	NX, Teamcenter	8
ME 2900/3900/4900	F23	J. Wagner, G. Mocko	NX, Teamcenter	7
ME 2900/3900/4900	S24	J. Wagner, G. Mocko	NX, Teamcenter	7
ME 2900/3900/4900	F24	J. Wagner, G. Mocko	NX, Teamcenter	8
ME 2900/3900/4900	S25	J. Wagner, G. Mocko	NX, Teamcenter	16
ME 3060: Fundamentals of Machine Design (Honors)	S24	Chris Mabey	SOLIDWORKS	2
ME 4170/6170: Mechatronic Systems Design	S19	John Wagner	Amesim	32
ME 4170/6170	S20	John Wagner	Amesim	30
ME 4170/6170	S21	John Wagner	Simcenter Amesim	30
ME 4170/6170	S23	John Wagner	Simcenter Amesim	36
ME 4170/6170	S24	John Wagner	Simcenter Amesim	26
ME 4710/6710: CAD Design & Analysis	F25	Satchit Ramnath	NX	34
ME 4930/6930: Mechatronics	S18	John Wagner	Simcenter Amesim	51
ME 4930/6930	S22	John Wagner	Simcenter Amesim	42
ME 4930/6930: Model Based Engineering	S21	Richard Figliola	Teamcenter	20
ME 4930/6930	F22	Cameron Turner	NX, Teamcenter	8
ME 4930/6930: PLM Processes	F23	J. Wagner, J. Morris	NX, Teamcenter	39
SAE Formula Student Team	F20	Greg Mocko	Star CCM+	10
SAE Formula	F23	Greg Mocko	Ansys	2
SAE Formula	F24	Greg Mocko	Ansys	24
Clemson Rocket Engineering Student Team	S23	Garrett Pataky	Onshape	16
AerosPACE Program	F18	Greg Mocko	CATIA	10
AerosPACE	F20	Greg Mocko	Onshape	5

Listing of Clemson University faculty using PLM Software:

Name	Department	Clemson Username	Telephone
Qiushi Chen	Civil Engineering	qiushi	864-656-3330
Matthew Krugh	Automotive Engineering	mkrugh	412-401-7764

Mary Beth Kurz	Industrial Engineering	mkurz	864-656-4652
James Forbes	Automotive Engineering	forbes6	864-283-7220
Kapil Madathil	Industrial Engineering	kmadath	864-656-0856
Laine Mears	Automotive Engineering	mears	864-283-7229
Greg Mocko	Mechanical Engineering	gmocko	864-656-1812
Garret Pataky	Mechanical Engineering	gpataky	864-656-2415
Chris Paredis	Automotive Engineering	paredis	404-981-6549
Robert Prucka	Automotive Engineering	rprucka	864-283-7223
Satchit Ramnath	Mechanical Engineering	sramnat	864-656-6248
Cameron Turner	Mechanical Engineering	cturne9	864-656-2413
John Wagner	Mechanical Engineering	jwagner	864-656-7376
Matthias Schmid	Automotive Engineering	schmid	864-283-7216

Appendix C. List of Seminars and Workshops Hosted by the PLM Center

Title	Date	Instructor	Audience	Students
PLM 2.3 Additive Man.	3/12/25	J. Morris	Seminar/Workshop	3
PLM 4.2 PDM	2/19/25	J. Morris	Seminar/Workshop	13
PLM 4.4 MBSE	2/5/25	J. Morris	Seminar/Workshop	11
PLM 4.1 Project Management	1/29/25	J. Morris	Seminar/Workshop	13
Advanced SOLIDWORKS	1/23/25	J. Morris	Seminar/Workshop	21
Intro to SOLIDWORKS	1/22/25	J. Morris	Seminar/Workshop	21
Intro to Siemens NX	1/17/25	J. Morris	Seminar/Workshop	38
PLM 2.4 Data Mining	11/13/24	J. Morris	Seminar/Workshop	4
PLM 4.3 Eng Change Mgmt.	11/6/24	J. Morris	Seminar/Workshop	4
PLM 3.3 GD&T	10/30/24	J. Morris	Seminar/Workshop	14
PLM 3.4 Sim and Opt.	10/23/24	J. Morris	Seminar/Workshop	9
Opportunities with PLM	10/17/24	J. Morris	ME 2000	410
Teamcenter for Beginners	9/13/24	F. Rowell	Seminar/Workshop	10
Introduction to PLM	9/5/24	J. Morris	ME 4020	264
PLM 1.1 Intro to PLM	9/4/24	J. Morris	Seminar/Workshop	25
PLM 2.3 Additive Manu.	3/5/24	J. Morris	Seminar/Workshop	7
PLM 4.1 Project Management	2/27/24	J. Morris	Seminar/Workshop	7
PLM 3.4 Sim and Opt.	2/20/24	J. Morris	Seminar/Workshop	14
4.4 MBSE	2/13/24	J. Morris	Seminar/Workshop	7
PLM 4.2 PDM	2/6/24	J. Morris	Seminar/Workshop	22
PLM 3.1 CAD Methods	1/30/24	J. Morris	Seminar/Workshop	5
Intro to Siemens NX	1/23/24	J. Morris	Seminar/Workshop	0
PLM 3.3 GD&T	11/14/23	J. Morris	Seminar/Workshop	5
PLM 2.3 Data Mining	11/7/23	J. Morris	Seminar/Workshop	3
PLM 2.2 Data Preservation	11/2/23	J. Morris	Seminar/Workshop	0
PLM 4.4 MBSE	10/24/23	J. Morris	Seminar/Workshop	6
SOLIDWORKS Intro	10/10/23	J. Morris	Seminar/Workshop	4
PLM 4.3 Eng Change Mgmt.	9/26/23	J. Morris	Seminar/Workshop	2
PLM 4.1 Project Management	9/19/23	J. Morris	Seminar/Workshop	5
PLM 3.1 CAD Methods	9/12/23	J. Morris	Seminar/Workshop	6
Introduction to NX	9/5/23	J. Morris	Seminar/Workshop	11
PLM 1.1 Intro to PLM	8/29/23	J. Morris	Seminar/Workshop	4
PLM 2.3 Additive Manu.	3/5/24	J. Morris	Seminar/Workshop	7
PLM 3.4 – Sys Sim Op	4/14/23	J. Morris	Seminar/Workshop	1
PLM 3.2 – CAE	4/7/23	J. Morris	Seminar/Workshop	0
PLM 2.3 – Add Manu.	3/31/23	J. Morris	Seminar/Workshop	1
PLM 2.4 – Data Mining	3/17/23	J. Morris	Seminar/Workshop	0
PLM 2.2 – Data Preservation	3/10/23	J. Morris	Seminar/Workshop	0
PLM 4.3 – Eng Change	3/3/23	J. Morris	Seminar/Workshop	0
PLM 2.1 – Product Des	2/24/23	J. Morris	Seminar/Workshop	2
SOLIDWORKS Intro	2/17/23	J. Morris	Seminar/Workshop	1
Teamcenter Training	2/10/23	J. Wagner, F. Rowell	DO 14/15 CU ICAR	50
PLM 3.3 – 2D Draw	2/10/23	J. Morris	Seminar/Workshop	2
PLM Concepts	2/9/23	J. Morris	ME 4020 Capstone	100
PLM Concepts	2/2/23	J. Morris	ME 2000 Seminar	70

PLM Concepts	2/1/23	J. Morris	ME 3120 Manu.	45
PLM Concepts	2/1/23	J. Morris	ME 3120 Manu.	40
NX Introduction	1/27/23	J. Morris	Seminar/Workshop	8
PLM 1.1 – PLM Intro	1/20/23	J. Morris	Seminar/Workshop	1
NX Training	S2023	J. Morris	Creative Inquiry	8
Teamcenter Training	S2023	J. Morris, F. Rowell	Creative Inquiry	8
PLM Concepts & Software	10/6/22	J. Morris	ME 4020 Capstone	66
PLM Center Overview	9/15/22	J. Morris	ME 2000	384
PLM 3.3 –GD&T	9/13/22	J. Morris	CUICAR ORD 255	2
PLM 3.2 – CAD Methods	9/7/22	J. Morris	PLM Lab – Clemson	3
PLM 3.2 – CAD Methods	9/6/22	J. Morris	CUICAR ORD 255	0
NX Training Sessions	F2022	J. Morris	Creative Inquiry	4
Introduction to NX	8/31/22	J. Morris	PLM Lab – Clemson	1
Introduction to NX	8/30/22	J. Morris	CUICAR ORD 255	3
Engineering Design	6/21-23/22	J. Wagner, J. Manning, J. Morris	PEER&WISE Camp	35
Simcenter Training Session	3/18/22	J. Wagner	ME 4930 & ME 8930	42
PLM Processes Workshop	4/20.22/22	J. Wagner	ME 4930 & ME 8930	42
PLM 2.3 – Additive Manu.	4/12/22	J. Morris	CUICAR ORD 203	0
PLM 2.2 – Data Preservation	3/30/22	J. Morris	PLM Lab – Clemson	3
PLM 2.2 – Data Preservation	3/29/22	J. Morris	CUICAR ORD 255	0
PLM 2.4 – Data Mine	3/10/22	J. Morris	PLM Lab – Clemson	1
PLM 2.4 – Data Mine	3/8/22	J. Morris	CUICAR ORD 225	0
PLM 4.1 – Fund PLM	3/1/22	J. Morris	CUICAR ORD 225	1
PLM 3.3 – 2D & GDT	2/17/22	J. Morris	PLM Lab – Clemson	2
PLM 3.3 – 2D & GDT	2/15/22	J. Morris	CUICAR ORD 225	11
PLM Concepts & Software	2/10/22	J. Morris	ME 4020 Capstone	138
PLM Center Overview	2/3/22	J. Morris	ME 2000 Seminar	55
NX Training Sessions	S2022	J. Morris	Creative Inquiry	5
Introduction to NX	2/2/22	J. Morris	PLM Lab – Clemson	1
Introduction to NX	2/1/22	J. Morris	CUICAR ORD 225	10
PLM 1.3 – Func. Areas	1/20/22	J. Morris	CUICAR ORD 225	5
PLM 1.3 – Func. Areas	1/19/22	J. Morris	PLM Lab – Clemson	0
Introduction to NX	11/11/21	J. Morris	CUICAR CGED 404	4
Introduction to NX	11/10/21	J. Morris	PLM Lab – Clemson	0
PLM 4.2 – PDM	11/4/21	J. Morris	CUICAR CGEC 404	0
Introduction to NX	11/3/21	J. Morris	PLM Lab – Clemson	1
NX Training Sessions	F2021	J. Morris	Creative Inquiry	7
PLM Center Overview	10/21/21	J. Morris	ME 2000 Seminar	111
PLM Concepts & Software	9/23/2021	J. Morris	ME 4020 Capstone	76
PLM 1.1 Workshop	9/7-9/21	J. Morris	352 EIB, CU ICAR	13
Engineering Design	6/21-24/21	J. Wagner, J. Manning	PEER&WISE Camp	55
PLM Center Overview	4/15/2021	J. Wagner	ME 2000 Seminar	75
PLM Processes Workshop	4/14/2021	J. Wagner	ME 4170/6170	30
Simcenter Training Session	2/10/2021	M. Kale	ME 4170/6170	30
PLM Concepts & Software	2/4/2021	J. Wagner	ME 4020 Capstone	135
PLM Concepts & Software	1/13/2021	J. Wagner	ME 4930/6930	20

PLM Center Overview	12/01/20	C. Wang	ME 2000 Seminar	250
CATIA Training Session	10/20/20, 10/23/20	C. Wang	PLMC Tutorial	2
NX Training Session	10/13-16/20	C. Wang	PLMC Tutorial	2
PLM Concepts & Software	9/24/20	C. Wang	ME 4020 Capstone	50
Siemens NX, 3DS Set up	3/13/20	E. Lambert	PLMC Seminar	3
3DE Simulation on Wheels	3/6/20	J. Wagner, E. Lambert	DS Product Offering	90
2D Drawings Part 1	3/5/20	E. Lambert	PLMC Seminar	12
Siemens NX, DS 3DE Set up	3/2/20	E. Lambert	PLMC Seminar	2
CAD Methods	2/27-28/20	E. Lambert	PLMC Seminar	21
DS 3DExperience / ENOVIA	2/26/20	E. Lambert	ME 4020 Capstone	6
Intro to Product Lifecycle	2/20/20	E. Lambert	PLMC Seminar	16
Siemens NX Teamcenter	2/19/20	Siemens	CU ICAR, PLMC	16
AMESim Workshop	2/11/20	S. Shoai-Naini	ME 4170/6170	29
DS 3DExperience / ENOVIA	2/11/20	E. Lambert	ME 4020 Capstone	3
PLM Concepts & Software	1/31/20	E. Lambert	Mechatronics Forum	15
PLM Center Overview	1/23/20	E. Lambert	ME 4020 Capstone	115
FEA Software Consultation	11/21/19	E. Lambert	Student Meeting	2
CAD and CAE Software	10/29-30/19,	E. Lambert	ME 4710/6710	56
PLM Center Overview	10/10/19	E. Lambert	PLMC Seminar	5
PLM Center Overview	10/1/19	E. Lambert	ME 4020 Capstone	50
PLM Software – NX, CATIA	9/4/2019	E. Lambert	ME 4710/6710	7
CAD Design & Innovation	6/25/19	E. Lambert	PEER & WISE Camp	71
3D Experience Onboarding	6/4-6/19	F. Salako	PLMC Laboratory	10
PLM Center Overview	6/3/19	E. Lambert	ME 4020 Capstone	30
PLM Center Overview	6/3/19	E. Lambert	ME 4010	30
NX Software – Features	05/22/19	E. Lambert	PLMC Laboratory	10
Introduction to PLM	05/21/19	E. Lambert	PLMC Seminar	10
3D Experience On Boarding	05/10/19	F. Salako	PLMC Laboratory	5

Appendix D. PLM Center Research (Publication, Projects)

Research Project Title	Dates	Investigators	PLM Software	No. Users
Representation Learning and Design Optimization in Engineering Applications	2023-	S. Ramnath	CATIA	3
Virtual Reality Toolchains for CAD Design	2022-25	G. Mocko, C. Turner	NX, Teamcenter	3
MSME Collaborative Design Enhancements Using Teamcenter	2023-	J. Wagner	NX, Teamcenter	45
Analysis of Hypergraphs for the Representation of Digital Twins	2024-	J. Wagner, G. Mocko	*N/A*	*N/A*
Development of Digital Twins for Radial Saws in Aluminum Cutting	2023-24	J. Wagner, G. Mocko	*N/A*	*N/A*
Telemedicine Systems and Compact Scanning Devices in Ambulances	2023-24	Kapil Madathil	NX	1
Exploration of Collaborative Design Spaces	2022-23	John Wagner	Teamcenter	1
Digital Twin of Boiler and HVAC System	2022-23	John Wagner	Simcenter Amesim	1
Clemson VIPR GS Center (US Army GVSC Funded Project)	2021-	Multiple	NX, Teamcenter	TBD
Digital Twin of Clemson University East Side Chiller Plant (Facilities Group)	2019-21	John Wagner	Simcenter Amesim	1
Lockheed Martin Add-In Software Evaluation	2020	J. Summers, J. Wagner	CATIA	2
Gas Turbine Turbulator Designs (GE Power)	2019-21	John Wagner	NX	1
Computational Geomechanics	2018-19	Qiushi Chen	Star CCM+	4
ARM Institute Quick Start	2018-19	V. Krovi, L. Mears	NX, Tecnomatix, Teamcenter, Jack	6
Functionally Designed Ultra-Lightweight Carbon Fiber Reinforced Thermoplastic Composites Door Assembly	2017-20	S. Pilla and G. Li	Tecnomatix, Fibersim	3
Dynamics Estimation and Control Lab	2023-	M. Schmid and L. Mears	Testlab	TBD

Appendix E. PLM Center Scholarly Publications

Book:

1. J. Morris, J. Wagner, and G. Mocko, "Product Lifecycle Management – Processes for a Modern Industry", September 2023 *(In progress)*

Journal Articles:

1. Eddy, Conner, Adam Wagner, John Wagner, and Matthew P. Castanier. 2022. "Application of a Digital Twin Virtual Engineering Tool for Ground Vehicle Maintenance Forecasting." *SAE International Journal of Advances and Current Practices in Mobility* 4 (4): 2022-01–0364. https://doi.org/10.4271/2022-01-0364.
2. Morris, John. 2025. "ConstraintHg: A Kernel for Systems Modeling and Simulation." *Under Review with Journal of Open-Source Software*, August.
3. Morris, John, Abhishek Indupally, Gregory Mocko, John Wagner, and Satchit Ramnath. 2025. "Declarative, Multi-Physics Simulation Between Applications via Constraint Hypergraphs." *Under Review with J. Comput. Inf. Sci. Eng.*, October.
4. Morris, John, Edward Louis, Douglas L. Van Bossuyt, Gregory Mocko, and John Wagner. 2025. "Constraint Hypergraphs as a Unifying Framework for Digital Twins." *Submitted to Nature Computational Science*, ahead of print, July. https://doi.org/10.48550/arXiv.2507.05494.
5. Morris, John, Gregory Mocko, and John Wagner. 2025a. "Effects of Functional and Declarative Modeling Frameworks on System Simulation." *Under Review with J. Dyn. Sys., Meas., Control (ASME)*, June.
6. Morris, John, Gregory Mocko, and John Wagner. 2025b. "Unified System Modeling and Simulation via Constraint Hypergraphs." *Journal of Computing and Information Science in Engineering* 25 (6): 061005. ASME Digital Collections (JCISE-24-1525). https://doi.org/10.1115/1.4068375.
7. Rowell, Frederick, John Morris, Todd Schweisinger, and John Wagner. 2025. "Digital Engineering Education Challenges and Opportunities–A Case Study in a Mechanical Engineering Senior Capstone Design Course." *Under Review with PLM Journal*, July.

Conference Articles:

1. Daly, Nicholas, Pranav Manvi, Tanmay Chhatbar, Matthias Schmid, Matthew P. Castanier, and John Wagner. 2024. *Modeling & Validation of a Digital Twin Tracked Vehicle*. SAE Technical Paper Nos. 2024-01–2323. SAE Technical Paper. https://doi.org/10.4271/2024-01-2323.
2. Eddy, Conner, Matthew Castanier, John Wagner, John Morris, and Benjamin Moss. 2023. "Usefulness and Time Savings Metrics to Evaluate Adoption of Digital Twin Technology." January 24, 14. https://doi.org/10.4271/2023-01-0111.
3. Eddy, Conner William, Matthew P. Castanier, and John R. Wagner. 2024. *Predictive Maintenance of a Ground Vehicle Using Digital Twin Technology*. SAE Technical Paper Nos. 2024-01–2867. SAE Technical Paper. https://doi.org/10.4271/2024-01-2867.
4. Manvi, Pranav, Darryl Suber Ii, Kaitlyn Griffith, Cameron Turner, Matthew P. Castanier, and John Wagner. 2025. *Validation of Product Requirements - Specifications Using Field Data and Next Generation Vehicle Design Strategy Using a Calibrated Digital Twin*. SAE Technical Paper Nos. 2025-01–8344. SAE Technical Paper. https://doi.org/10.4271/2025-01-8344.

5. Morris, John, Gregory Mocko, and John Wagner. 2025. "Effects of Functional and Declarative Modeling Frameworks on System Simulation." Paper presented at The 5th Modeling, Estimation and Control Conference (MECC 2025), Pittsburgh, PA. October.
6. Morris, John, Gregory Mocko, John Wagner, and Satchit Ramnath. 2025. "Declarative Integration of CAD Software into Multi-Physics Simulation via Constraint Hypergraphs." Paper presented at ASME IDETC-CIE 2025, Anaheim, CA. *Proceedings of the ASME 2025 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, August 17.
7. Morris, John, Dave Procopio, and John Wagner. 2023. "Dynamic Estimator of CAD Patterning Feature Execution Time." Paper presented at ASME IDETC-CIE 2023, Boston, MA. *Proceedings of the ASME 2023 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, November 21. https://github.com/Clemson-PLMC/PatternEstimatorTool.
8. Morris, John, and John R. Wagner. 2023. "Application of Extracurricular Course Teaching Product Lifecycle Management Concepts to Undergraduates." Paper presented at 2023 ASEE Annual Conference & Exposition, Baltimore, MD. June 25. https://doi.org/10.18260/1-2--42269.
9. Procopio, David, John Morris, and John Wagner. 2023. "Evaluation of CAE Design Change Updates--A Case Study on Gas Turbine Airflow Disruptors." Paper presented at ASME IDETC-CIE 2023. *Proceedings of the ASME 2023 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, November 21. https://doi.org/10.1115/DETC2023-116761.
10. Procopio, David, John Morris, and John R. Wagner. 2023. "Application of Product Lifecycle Management in the University Classroom and Laboratory." Paper presented at 2023 ASEE Annual Conference & Exposition, Baltimore, MD. June 25. https://doi.org/10.18260/1-2--42669.
11. Rowell, Frederick, Todd Schweisinger, and John Wagner. 2024. "Exploration of Collaborative Design Spaces: Student Engineering Interactions and Workflows in Product Development." Paper presented at ASEE Southeast Section Conference, Starkville, USA. November 11.
12. Rowell, Frederick, Todd Schweisinger, and John Wagner. 2025. "Impact of PLM Tools and Strategies on a Senior Capstone Design Project - Collaboration, Creativity, and Design Time." Paper presented at 2025 ASEE Annual Conference, Montreal, Canada. June 22.

Posters and Presentations:

1. Ainsworth, William, Brady Bostic, Jake Czawkiel, et al. 2025. "Development of Mechanisms Supporting Digital Twins in a Model-Based Engineering Ecosystem." Poster. With John Wagner, Gregory Mocko, and John Morris. Clemson University 20th Annual Focus on Creative Inquiry Forum, Clemson, SC, April.
2. Anderson, Lia, William Bishop, Brandon Ewanick, et al. 2023. "Exploring Digital Technologies - A Focus on PLM Concepts and Software with Application to Scale Track Vehicle." Poster. 18th Annual Focus on Creative Inquiry Forum, Clemson, SC, April. https://ci.clemson.edu/foci/posters2023/.
3. Anderson, Lia, Clarke Coco, Patrick O'Neill, et al. 2024. "Development of Digital Twin for Scaled Robotic Vehicle." Poster. 19th Annual Focus on Creative Inquiry Forum, Clemson, SC, April 3. https://ci.clemson.edu/foci/posters/.

4. Beuckman, Clark, Michael Calamari, Sam Gossett, Noah Wanthal, and Brenden Schumm. 2022. "PLM Processes and CAD/CAE Tools with Application to Vehicle Component Design." Poster. With John Morris, Gregory Mocko, and John Wagner. 17th Annual Focus on Creative Inquiry Forum, Clemson, SC, April.
5. Morris, John. 2024. "Representing Digital Twins." ASME IDETC-CIE: SciTechBuzz Summit 2024, Washington DC, August 26.
6. Morris, John. 2025a. "Model-Based Engineering of Digital Twins." Poster. VIPR-GS Research Center Annual Review Meeting, Greenville, SC, February 26.
7. Morris, John. 2025b. "Solving Interoperability with Digital Twins." Invited Presentation. Guest Presentation to the National Institute for Standards and Technology, Virtual, April 16.
8. Morris, John. 2025c. "Universal System Simulation via Hypergraphs." Applied Category Theory Conference (ACT 2025), Gainesville, FL, June 2.

PLM Center Directed Research Projects:

1. Joarder, M., (PhD ME), "Integration of Engineering Applications for Construction of Next Generation Digital Twins", (May 2029).
    a. Research ways of communicating PLM information between CAx applications to support the construction of digital twins for use in Industry 4.0.

2. Morris, J. (PhD ME), "Universal Systems Simulation via Constraint Hypergraphs with Applications to Digital Twins", (Dec 2025).
    a. Development of constraint hypergraphs, a novel modeling framework that enables declarative simulation of generalized systems, and provides a mathematical foundation for digital twins.

3. Kolangarakath, A., (MS IE), "Telemedicine Systems and Compact Scanning Devices in Ambulances", (May 2024).
    a. Research project with advisor Dr. Kapil Madathil that will use Siemens NX software to design and develop 3D models of an ambulance, stretcher, and scanners.

4. Rowell, F., "Application of Siemens Team Center in Engineering Collaborative Projects at Clemson University", *Honors Thesis*, (May 2023).
    a. Senior mechanical engineering student with experience using TeamCenter from two internships at E-Z-Go (Augusta, GA) will explore its application in Creative Inquiry (ME 2900/3900/4900) and Deep Orange (AuE Department, CU ICAR) team settings.

5. Kale, M., (MS ME), "Development and Application of a Digital Twin for Chiller Plant Performance Assessment", (December 2021).
    a. A graduate research project with Dr. Wagner, Clemson University Facilities Group, and Optimum Energy (Seattle, WA). He created a digital twin of the East Side Chiller Plant on the Clemson, SC campus using Siemens SimCenter AMESIM. The digital twin features four-loops - refrigeration (chiller), condenser water (cooling towers), chilled water (evaporator), and building side air. It included components such as compressors, pumps, heat exchangers, and fans. SimCenter AMESIM

provides graphs to verify and validate against the available monitoring data collected daily by the Facilities Group. The goal was to create a tool to help in the monitoring and predictive maintenance of these HVAC plants.

6. Procopio, D., (MS ME), "Product Lifecycle Management – Application of Patterning Methods to Gas Turbine Blades and Creation of Learning Materials", (December 2021).
    a. An advanced CAD research project with General Electric (GE Power) and the PLM Center to evaluate the time requirements for design changes in automatic software generated patterns. He explored the amount of time required for model modifications to cascade through a complex component CAD description once it was changed from the original state. Using Siemens NX and gas turbine turbulator modeling (surface discontinuities inserted inside blades to promoted heat transfer with air flow disruptions), the automatic pattern generators and blade geometry processing time were studied. In addition, he helped to create learning materials for the K-12 outreach efforts including the 1-page assessment survey to gather student feedback.

Appendix F. Overview of Developed PLM Modules

PLM-1.0: Fundamental Concepts

1.1 Introduction to the Produce Lifecycle

An introduction to Product Lifecycle Management with coverage of the fundamental concepts. Insight given on the development of PLM strategies and evolution to the current practices. Each phase of process is described with examples of the actions, activities, and ideas that occur within an organization.

1.2 Digital Components of PLM

A general overview of computer-aided-design (CAD), -engineering (CAE), and -manufacturing (CAM) as well as engineering data management (EDM) and digital manufacturing (DM) design capabilities. The topics of PLM data and accompanying product data management (PDM) tools plus enterprise resource planning (ERP) are introduced.

1.3 PLM Functional Areas

Opportunities to collaborate across the business enterprise are reviewed using PLM concepts within a standard configuration. Companies organize about core skills (functional areas including product management, engineering, manufacturing, supply chain, support) which are reviewed, and their interactions highlighted during production.

1.4 PLM in Practice

An example product is developed using PLM concepts to demonstrate the process. The PLM functional areas are discussed with the tasks of product conception, development, realization, field service, and end-of-life. Opportunity exists for student identified virtual product applications to work through the various tasks.

PLM-2.0: Digital Product Development

2.1 Product Design Tools and Approaches

One challenge when adopting PLM tools is the reorganization of existing engineering processes to align with the digital tools and standardize the workflow. Through a case study, the primary steps and possible pitfalls when implementing a digital process are discussed.

2.2 Data Preservation and Security

The rigorous saving of digital data to preserve design data as well as the implementation of computer security and information protocols must be considered when transitioning to a digital workflow process. Best practices and accompanying challenges will be discussed with attention focused on working with information technology specialists.

2.3 Additive Manufacturing

The growth in 3D printing and other additive manufacturing processes can support the rapid transition from virtual to physical prototype. The output of CAD/CAE computer programs (STL – Standard Triangulation/Tessellation Language) can be directly interfaced to additive printers to create layer-by-layer polymer, metal, composite parts. This course explores the transition from digital to actual components with attention focused on common processes.

2.4 Data Mining and Analytics
Big data and product lifecycle management – what is the connection? In this course, the raw data generated from digital simulations, product field operation, and other sources will be statistically analyzed to identify trends, patterns, and other important features.

PLM-3.0: Graphical Concepts

3.1 Computer Aided Design Methods (CAD)
Insight into design for excellence and lean product development by explaining strategies for computer-aided-design (CAD) models and assembly reuse. The course also covers additional assembly and modeling concepts such as modeling in space and associate modeling.

3.2 Computer Aided Engineering (Design / Assembly)
An introduction into computer-aided-engineering (CAE) methods with focus on efficient creation and reuse of simulations. Explanation of common product development practices and how PLM technology allows for more developer control through concurrent design.

3.3 Two Dimensional Drawings and Geometric Dimensioning / Tolerancing
A graphics primer course that examines the concepts of form, fit and function (FFF). The importance of FFF and how it is used to specify product with a large emphasis on geometric dimension and tolerancing (GD&T) are covered through theory and example.

3.4 System Simulation and Optimization
A sequential demonstration of simulation and optimization graphical concepts and tools useful in research projects. The discussion focuses on multi-disciplinary systems including the integration of electro-mechanical components into complex assemblies.

PLM-4.0: Non-Graphical Concepts

4.1 Project Management
A review of project management activities with an emphasis on project initiation using key administrative concepts. The workflow process includes product scope, requirements, work breakdown structure, cost, schedules, implementation strategy, and risk assessment.

4.2 Product Data Management
The basics of product data management (PDM) and repository development are covered in this course. The topics include product structure, configuration management, requirements modeling, SysML, analysis data management, interoperability, and data exchange.

4.3 Engineering Change Management
Students will learn the sequence of tasks to be completed to drive improvements in a product or work process. The workflow for engineering change will be discussed and examples offered on how changes in engineering information may propagate through the engineering database.

4.4 Model Based Systems Engineering (MBSE)
The creation of computer models and supporting virtual framework to support the design, analysis and management of complex physical systems through requirements, simulation & analysis, systems integration, verification & validation, and change management.

PLM Software Training and Help Sessions

Siemens NX Software – Basic Features with Application
Learn how to use select features of the NX software package and apply this computer-aided-engineering tool to design a component on the PLM Center Computer Laboratory machines.

Dassault Systèmes Software – Introduction and Applications
PLM software package used for three-dimensional design. Learn some of the digital engineering tool's primary features and practice on creating virtual parts.

Siemens Teamcenter Software - Introduction and Basic Features
Fundamental concepts in Teamcenter shared in slide package; ME H4150 deliverable.

Appendix G. Student Assessment Survey (IRB Approved)

Evaluation of PLM Center at Clemson University Sponsored Workshops and Seminars

Please carefully consider each of the following questions while considering only the most recently attended workshop / seminar. The estimated time to take this survey is less than 5 minutes.

Question 1 - The concepts in the workshop / seminar were interesting to me.
1) Strongly Disagree, 2) Disagree, 3) Neither Agree nor Disagree, 4) Agree, 5) Strongly Agree

Question 2 - The workshop / seminar has motivated me to continue learning about PLM topics.
1) Strongly Disagree, 2) Disagree, 3) Neither Agree nor Disagree, 4) Agree, 5) Strongly Agree

Question 3 - The workshop / seminar was well-organized, and it clearly explained digital concepts.
1) Strongly Disagree, 2) Disagree, 3) Neither Agree nor Disagree, 4) Agree, 5) Strongly Agree

Question 4 - The workshop / seminar clearly established the relevance of the topics to business and industry.
1) Strongly Disagree, 2) Disagree, 3) Neither Agree nor Disagree, 4) Agree, 5) Strongly Agree

Question 5 - The knowledge gained in this workshop / seminar will positively impact my (future) academic studies and/or professional career.
1) Strongly Disagree, 2) Disagree, 3) Neither Agree nor Disagree, 4) Agree, 5) Strongly Agree

Question 6 - Which of the following would have the greatest impact in making the workshop / seminar more effective? (Multiple responses may be selected)

- ☐ Revise the presentation content.
- ☐ Improve the presenter's delivery and/or communication skills.
- ☐ Provide more examples, demonstrations, and/or walkthroughs of digital concepts.
- ☐ Offer better access to PLM software and/or learning resources.

Question 7 – What interested you in attending this event?
(Free Response)

Question 8 - Please share any suggestions you have about the workshop/seminar, the PLM Center, PLM software tools and processes, or future offerings.
(Free response)

Question 9 - Would you recommend this workshop / seminar to a friend or colleague (Yes/No)? Why?

- ○ Yes
- ○ No

(Free response)

Question 10 – What is your academic/professional status?

- ○ Freshman

- Sophomore
- Junior
- Senior
- Graduate student
- Clemson University faculty/staff
- Technical professional
- Other

Thank you for your time and responses. If you have any questions or concerns, please reach out to the Product Lifecycle Management Center at plmc@clemson.edu.

Your responses will be part of an ongoing research study being conducted by the PLM Center directed by Dr. John Wagner. Specific questions about this research can be addressed to him at jwagner@clemson.edu.

Appendix H. Table of Contents of PLM Concepts Textbook

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Figure H1: Screenshots of Table of Contents from PLM Concepts textbook. Seven chapters have been drafted, with three more still to be completed.